

Rockborne

Ridge Regression

Taming coefficients when features collide

Objectives



Ridge Defined

What it is and why it exists.



Use Case & Benefits

When and why to use it.



From SLR to Ridge

A conceptual walkthrough.



Equations & Walkthrough

The L2 penalty.

Introduction



Intro to Ridge Regression

What is Ridge Regression?

What is it?

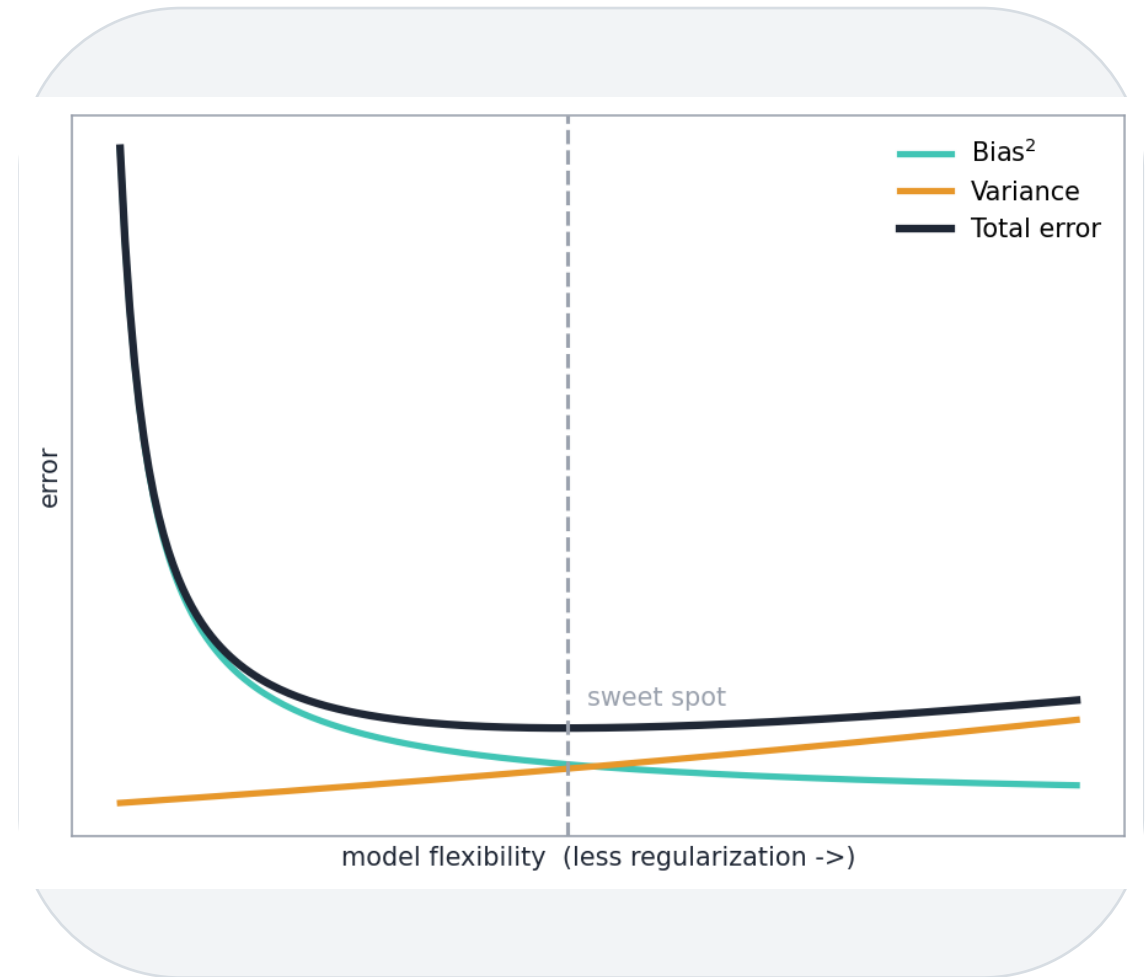
- A regression method for when multicollinearity is present.
- Ordinary least squares minimises the sum of squared residuals (RSS).
- Ridge adds a penalty on the size of the coefficients.

When do we use it?

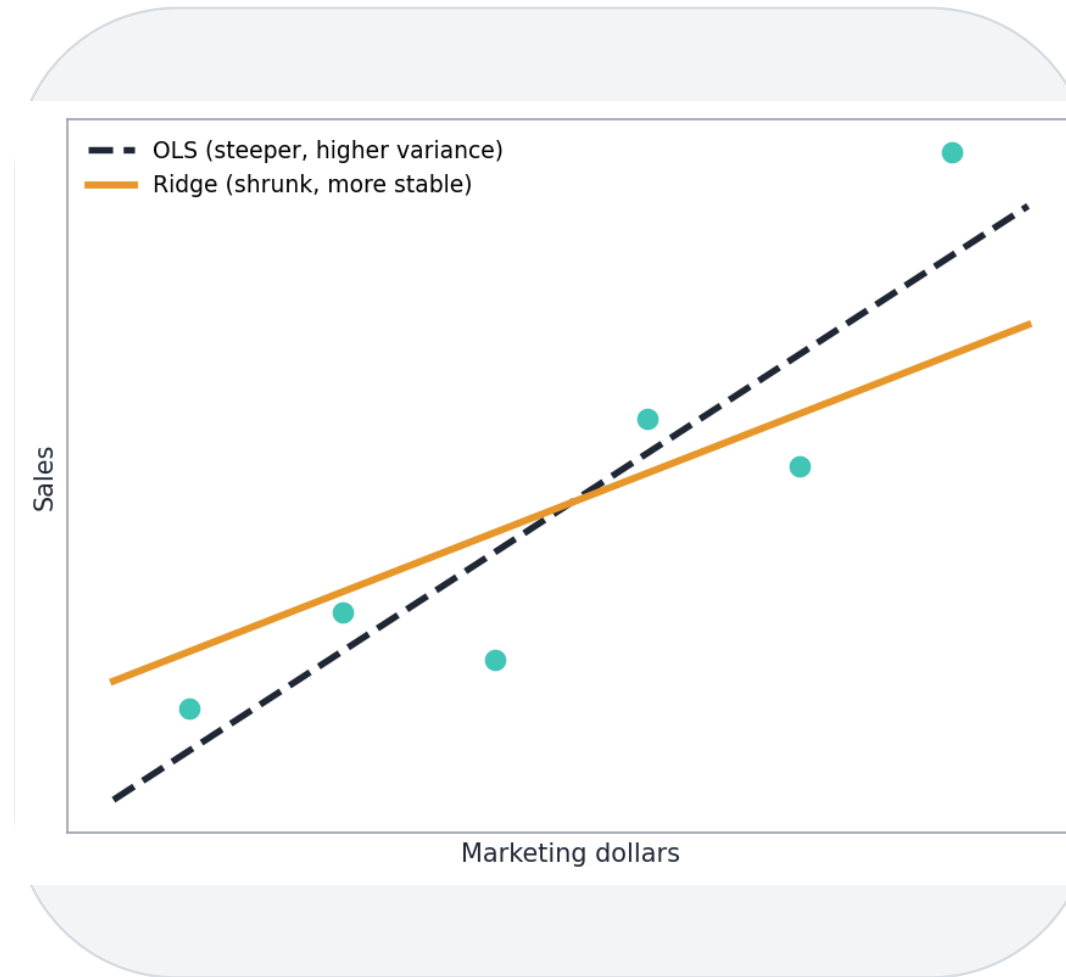
- When predictor variables are highly correlated.
- Multicollinearity makes coefficient estimates unreliable.
- It reduces their high variance.

Benefits

- Increases the flexibility of linear models by trading a little bias.
- Introducing a small bias substantially reduces variance.
- That leads to a lower overall MSE.
- The goal is the sweet spot between bias and variance.



Conceptual Example: from SLR to Ridge



1

Simple regression can overfit a few noisy points (a steep line).

2

Ridge shrinks the slope for a more stable fit.

3

Example: predicting sales from marketing dollars.

4

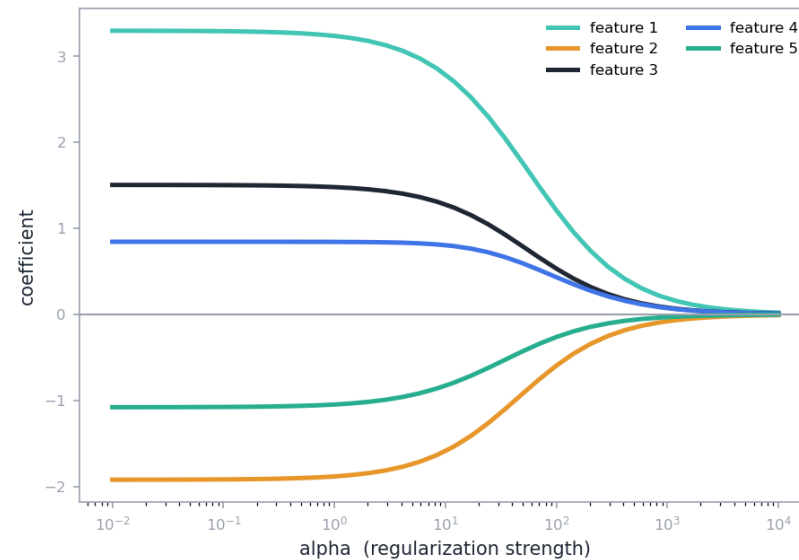
A little bias now, better predictions on new data.

Ridge Regression



Equations and Walkthrough

The Ridge Equation



OLS minimises RSS (the sum of squared residuals).

Ridge minimises $\text{RSS} + \alpha \times (\text{sum of squared coefficients})$ - the L2 penalty.

α controls the strength: 0 = OLS, larger = more shrinkage.

Coefficients shrink toward zero (never exactly zero).
Scale features first.

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Time to get hands-on in Python

We move from theory to practice with a guided walkthrough you run yourself in Python.



Ridge Regression - Guided Walkthrough



Then: Lasso Regression

In this lesson, we explored



Intro to Ridge

A fix for multicollinearity and unstable coefficients.



Benefits

Trade a little bias for much less variance.



The Equation

RSS + an L2 penalty, controlled by alpha.

Digging Deeper?



[Ridge Regression in Python](#)

Statology



[Ridge \(API\)](#)

scikit-learn



[Ridge Regression \(User Guide\)](#)

scikit-learn

Questions?